

city	country	salary
Bangalore	India	18
Abu Dhabi	UAE	80
Hyderabad	India	16
Dubai	UAE	80
Bangalore	India	20

Direct extraction (low-dimensional)

Condition
 $Q=0$
 $0 < Q < 1$
 $Q=1$

Implication
 Functional
 Logical
 No dependency

Dependency
extraction
tool

Hyperedge-wise
extraction

Logical dependencies

$\{ \text{"city"} \} \sim \{ \text{"salary"} \} (q=0.0833)$

$\{ \text{"salary"} \} \sim \{ \text{"city"} \} (q=0.0833)$

Functional dependencies

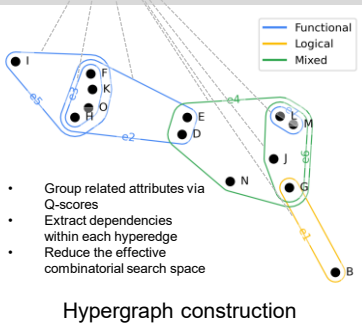
$\{ \text{"city"} \} \rightarrow \{ \text{"country"} \}$

Note: Functional dependencies are a special case of logical dependencies

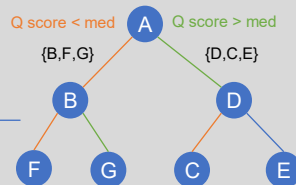
Hypergraph guided extraction
(high-dimensional)

A		Z
Bangalore		18
Abu Dhabi		80
Hyderabad		16
Dubai		80
Bangalore		20

Computing dependency matrix
using brute-force method



Hypergraph construction for
very high-dimensional data



Feature tree with KNN search fastens the computation of dependency matrix

Computing dependency matrix
using tree-method